departments at Mines but still desire graduate training in humanitarian engineering and science. It consists of four courses.

In both the master's degrees and graduate certificates, a unique mix of social science, applied science, and engineering perspectives prepares students to apply knowledge about the earth to promote more sustainable and just uses of water, energy, and other earth resources and to understand and mitigate potential hazards.

Primary Contact

Program Director: Richard Krahenbuhl
Program Coordinator: Rachel McDonald
humanitarian.mines.edu/mshes/

Graduate Certificate Program Requirements

The *Humanitarian Engineering and Science (HES)* certificate is an online or residential program designed for working professionals as well as graduate students who are enrolled in other degrees at Mines but wish to gain knowledge in humanitarian engineering and science. To obtain a graduate certificate, students must complete a minimum of 9 credits of the following courses. Students may not double-count courses from their undergraduate degrees. Students who have already taken one of the classes as undergraduates must find a suitable replacement, to be approved by the HES director. Students are encouraged to take 12 credits of coursework if possible, adding an elective from the approved HES electives list below.

Required HES certificate courses (9 credits):

EDNS515	INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES	3.0
EDNS577	ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT	3.0
EDNS579	COMMUNITY-BASED RESEARCH METHODS	3.0
Total Semester Hrs		**9.0**

Master of Science (MS) Program Requirements

The MS degrees in *Humanitarian Engineering and Science (HES)* are a professional MS (non-thesis) and a thesis-based MS. These degrees are targeted to recent graduates or midcareer professionals with a BS in science and engineering who are interested in careers, research opportunities, and/or acquiring skills that will help them work effectively with communities. The degrees include a core HES curriculum plus an approved track of related courses in a science or engineering discipline. A unique mix of social science, applied science, and engineering perspectives prepares students to apply knowledge about the earth to promote more sustainable and just uses of water, energy, and other earth resources and to understand and mitigate potential hazards.

Master of Science (non-thesis)

To obtain the 30 credits required for the MS (non-thesis), students must satisfy the following program requirements: (1) 12 credits of required HES courses, 2) 3 credits of elective HES courses approved by Engineering, Design & Society, and 3) 15 credits of courses approved by the affiliated department (see the six tracks detailed below).

HES MS (Non-Thesis) Core Courses (15 credits):

EDNS515	INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES	3.0
EDNS577	ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT	3.0
EDNS579	COMMUNITY-BASED RESEARCH METHODS	3.0
EDNS580	HUMANITARIAN ENGINEERING AND SCIENCE CAPSTONE PRACTICUM	3.0
ELECTIVE	An approved HES elective from the list below	3.0
Total Semester Hrs		**15.0**

Approved HES Electives:

EDNS590	RISKS IN HUMANITARIAN ENGINEERING AND SCIENCE	3.0
CEEN501	LIFE CYCLE ASSESSMENT	3.0
CEEN575	HAZARDOUS WASTE SITE REMEDIATION	3.0
CEEN593	SUSTAINABLE ENGINEERING DESIGN	3.0
CEEN556	MINING AND THE ENVIRONMENT	3.0
CEEN570	WATER AND WASTEWATER TREATMENT	3.0
CEEN573	RECLAMATION OF DISTURBED LANDS	3.0
CEEN580	CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT	3.0
CEEN581	WATERSHED SYSTEMS MODELING	3.0
CEEN595	ANALYSIS OF ENVIRONMENTAL IMPACT	3.0
EBGN553	PROJECT MANAGEMENT	3.0
HASS525	ENVIRONMENTAL COMMUNICATION	3.0
HASS526	INTERCULTURAL COMMUNICATION	3.0
HASS527	RISK COMMUNICATION	3.0
HASS565	SCIENCE, TECHNOLOGY, AND SOCIETY	3.0
HASS568	ENVIRONMENTAL JUSTICE	3.0
HASS590	ENERGY AND SOCIETY	3.0
MNGN503	MINING TECHNOLOGY FOR SUSTAINABLE DEVELOPMENT	3.0
MNGN510	FUNDAMENTALS OF MINING AND MINERAL RESOURCE DEVELOPMENT	3.0
MNGN565	MINE RISK MANAGEMENT	3.0
MNGN567	SUSTAINABLE DEVELOPMENT AND EARTH RESOURCES	3.0
MNGN571	ENERGY, NATURAL RESOURCES, AND SOCIETY	3.0
PEGN530	ENVIRONMENTAL LAW AND SUSTAINABILITY	3.0

Disciplinary Tracks

Track 1: Geophysics (GPGN) (15 credits):

Degree candidates should have an undergraduate degree in geophysics, physics, quantitative earth sciences and engineering, or equivalent coursework. In addition, candidates will need to complete necessary prerequisite courses for the graduate courses.

In addition to the core HES MS (non-thesis) curriculum (15 credits) detailed above, MS (non-thesis) students following the Geophysics track must take one required course (3 credits) and at least 12 credits of approved elective courses, as shown below. Courses not listed below that align with the student's practicum can be substituted in consultation with the degree advisor.

Required Course		
GPGN577	HUMANITARIAN GEOSCIENCE	3.0